

Lecture 11: Tableau-style proofs

AI504 - Knowledge Representation

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April/2026



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1 Definition

Definition of an analytic tableau

A tableau (for propositional logic) is a tree where each node is labeled by two sets of formulas (a T -set and a F -set) and each non-leaf node has an active connective, which is the topmost connective in on one the formulas subject to the following rules:

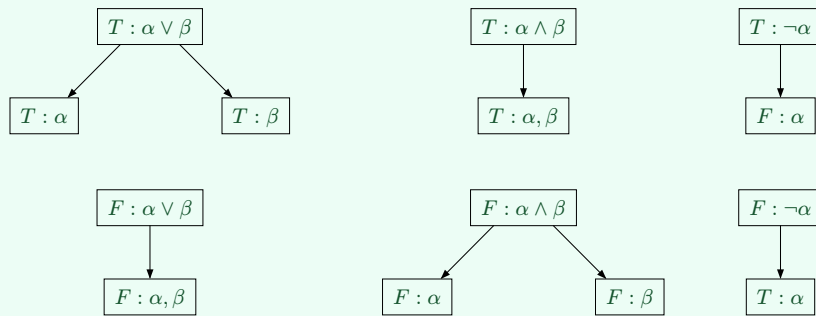


Figure 1: Rules for tableau expansion (propositional logic)

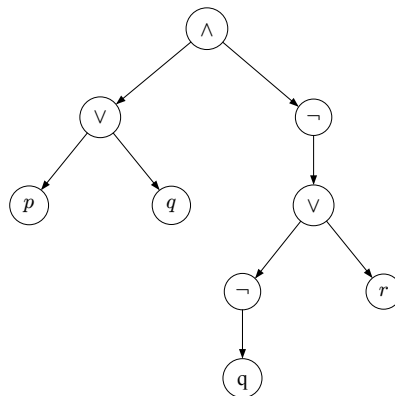
2 Example

Consider the following formula

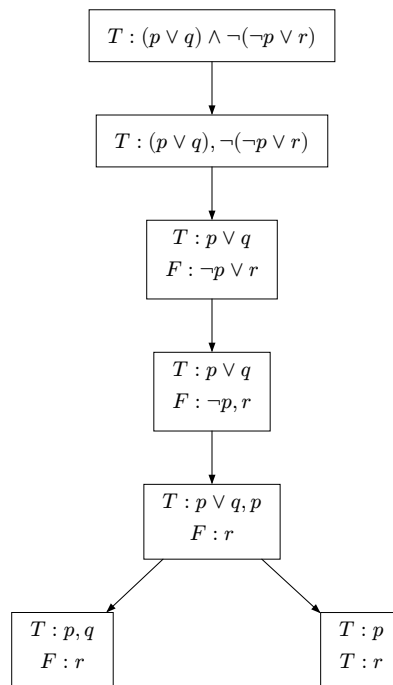
$$(p \vee q) \wedge \neg(\neg p \vee r)$$

2.1 Can we find a satisfying model $\mathcal{M}$ for this formula?

Lets see this as the following tree.



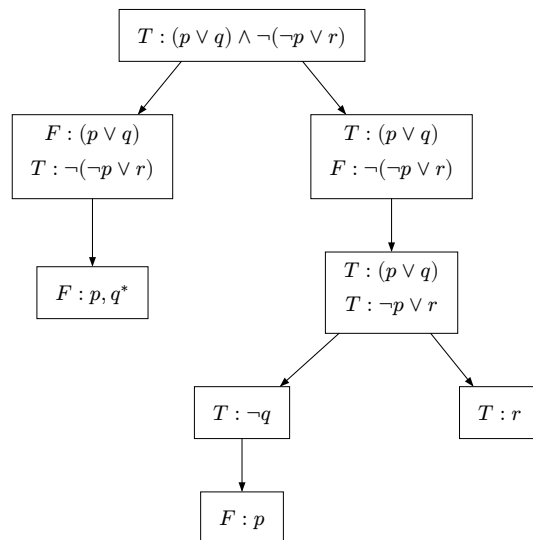
Write this as a tableau-style proof. We start with the root node as $T : (p \vee q) \wedge \neg(\neg p \vee r)$



Both of the last nodes are leaves.

2.2 Can we a falsifying model $\mathcal{M}$ for this formula?

Write this as a tableau-style proof. We start with the root node as $T : (p \vee q) \wedge \neg(\neg p \vee r)$



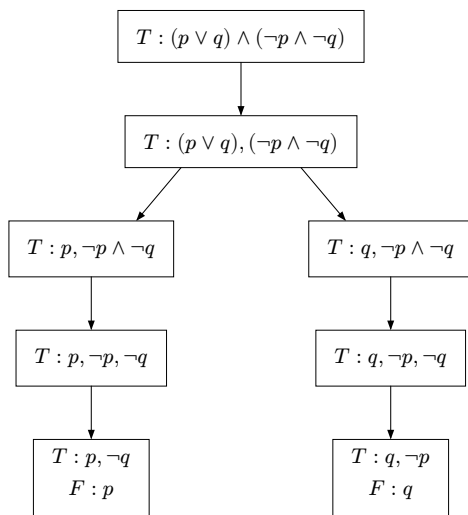
Note that $*$ is enough for a falsifying model

2.3 What if a model $\mathcal{M}$ does not exists?

New sentence

$$(p \vee q) \wedge (\neg p \wedge \neg q)$$

This sentence is unsatisfiable and a contradiction. We can show this by a tableau-style proof.



These leaves are obvious contradictions. But to show that there exists no model, we need to show that all branches result in a contradiction.

3 Check formula x

3.1 Satisfiability or contradiction

To check if a formula x is satisfiable, we can start with the root node as $T : x$ and apply the tableau expansion rules until we reach the leaves. If all branches end in a contradiction, then x is unsatisfiable. If at least one branch does not end in a contradiction, then x is satisfiable by a model $\mathcal{M}$.

3.2 Falsifiability or tautology

Note that for x to be satisfiable, does not mean that x is not falsifiable. If there exists no falsifying model for x , then x is a tautology.

4 Recall the inference problem

Given a set of sentences Γ and a single sentence ϕ , either produce a proof of ϕ from Γ (whatever that might be), or a model satisfying Γ but falsifying ϕ

To determine whether $\Gamma \models \phi$:

Look for a model (via the method of tableaux) on the form

$$\begin{array}{l} T : \Gamma \\ F : \phi \\ \vdots \end{array}$$

If we find such a model in a leaf, we have our counter-model, & this means that $\Gamma \not\models \phi$

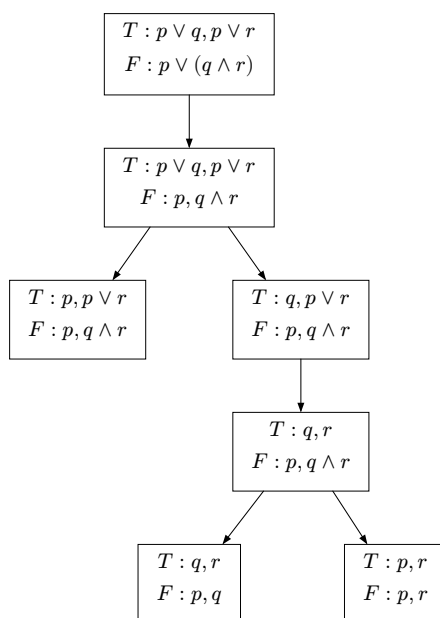
If on the other hand every one of our leaves ends up in a contradiction, this counts as a proof that $\Gamma \models \phi$.

Correctness of tableau method gives us a decision procedure for inference problem of propositional logic.

5 Example

Let $\Gamma = \{p \vee q, p \vee r\}$ and $\phi = p \vee (q \wedge r)$

Does $\Gamma \models \phi$



Since all leaves are contradictions, this counts as a proof of

$$\{p \vee q, p \vee r\} \vdash p \vee (q \wedge r)$$